

Official-source Electricity compilation 08

Mixed-source chapter worksheet • cross-paper reuse disclosed

Total: 33 marks • Suggested time: 66 minutes

Instructions and reuse disclosure

Curated compilation of official CBSE source questions, not a CBSE-issued chapter paper. Original source labels and exact source images are retained.

Answer every selected question. Only one selected source-slot alternative appears in this paper; do not answer or count excluded alternatives.

Questions recur across this 15-paper collection. Reuse is disclosed per question in the source appendix; these are not 15 disjoint official papers.

Sections use available source marks: A MCQ; B 2 marks; C 3 marks; D 5 marks or source-marked case questions. Case marks are not inflated to five.

Section A — MCQ

Compilation questions 1-4; 4 marks. Exact source labels remain attached to each question.

Section B — 2-mark questions

Compilation questions 5-8; 8 marks. Exact source labels remain attached to each question.

Section C — 3-mark questions

Compilation questions 9-12; 12 marks. Exact source labels remain attached to each question.

Section D — 5-mark / case-based questions

Compilation questions 13-14; 9 marks. Exact source labels remain attached to each question.

Section A — MCQ

1. [1 marks] Source Q31 • 2026 First Board

- 31.** In an electric circuit carrying current I , a $2\ \Omega$ resistor is replaced by an $8\ \Omega$ resistor without changing the connected (ideal) battery. Then : 1
- (A) current becomes $\frac{I}{4}$ and potential difference $\frac{V}{4}$
- (B) current becomes $\frac{I}{4}$ and potential difference remains V
- (C) current becomes $4I$ and potential difference remains V
- (D) current becomes $4I$ and potential difference $\frac{V}{4}$

31/3/2

Page 23 of 31

P.T.O.

Source/editorial note: Visually confirmed complete replacement crop: question number, wording, all options/table values, mathematical symbols and printed marks. Whitespace normalized; mathematical glyphs represented in Unicode and stacked fractions as $I/4$ and $V/4$. Page/QR/footer debris removed from text; marks retained in marks field. No supplied diagram; table retained in text and evidence. Answers not reviewed. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

2. [1 marks] Source Q31 • 2026-27

31	An electrical appliance rated 1000 W is operated at 250 V. What current does it draw?	1
	A. 0.25 A	
	B. 2.5 A	
	C. 4.0 A	
	D. 10 A	

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

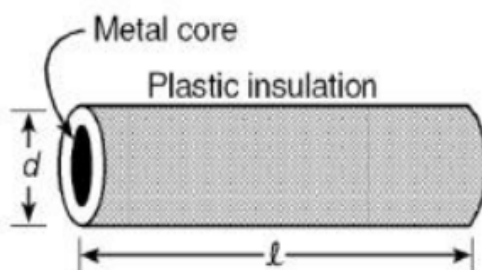
3. [1 marks] Source Q7 • 2019-20

- 7 Unit of electric power may also be expressed as: 1
- i) volt-ampere
- ii) kilowatt-hour
- iii) watt-second
- iv) joule-second

4. [1 marks] Source Q15 • 2022-23

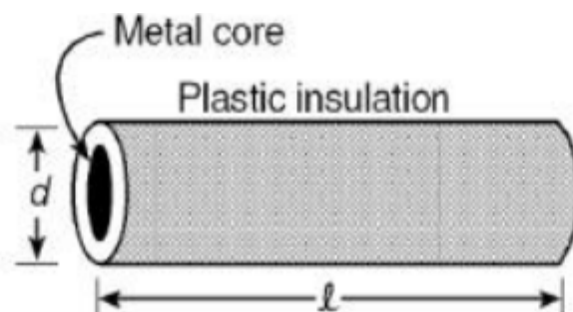
15

1



Plastic insulation surrounds a wire having diameter d and length l as shown above. A decrease in the resistance of the wire would be produced by an increase in the

- (a) length l of the wire
- (b) diameter d of the wire
- (c) temperature of the wire
- (d) thickness of the plastic insulation

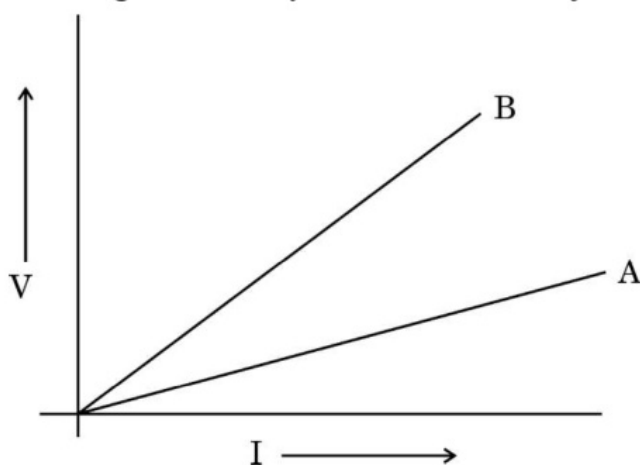


Section B — 2-mark questions

5. [2 marks] Source Q25 • main

- 25.** $V - I$ graph for two conducting wires A and B are as shown. If both wires are of the same length and same diameter, which of the two is made of a material of high resistivity? Give reasons to justify your answer.

2



Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

6. [2 marks] Source Q26 • main

- 26.** Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

2

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

7. [2 marks] Source Q25 VI B • 2024-25

OR

- B. You are given 2 fuse wires A and B with current ratings 2A and 5A respectively. Justify with reason which of the two would you use with a 1000W, 220V room heater?

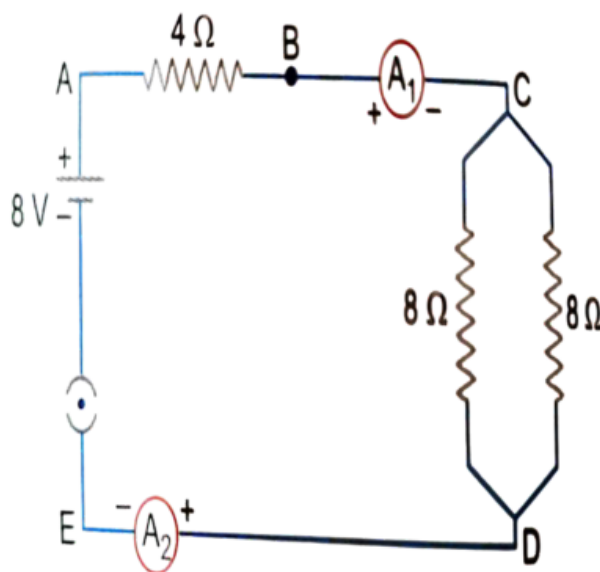
Source/editorial note: Borderline household safety context retained by explicit scope instruction: electric-power/current calculation and fuse rating. Original MS wording “with is” retained. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

8. [2 marks] Source Q34 A • 2025-26

34

Attempt either option A or B.

A.



Find out the following in the electric circuit given in the figure-

- Effective resistance of two 8 ohm resistors in the combination.
- Current flowing through the 4-ohm resistor

2

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section C — 3-mark questions

9. [3 marks] Source Q27 • main

27. (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is $14\ \Omega$. If the resistivity of the material of the wire is $44 \times 10^{-8}\ \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)

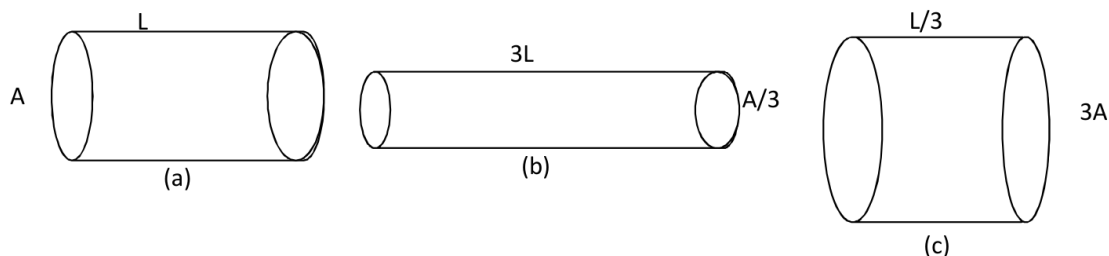
3

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

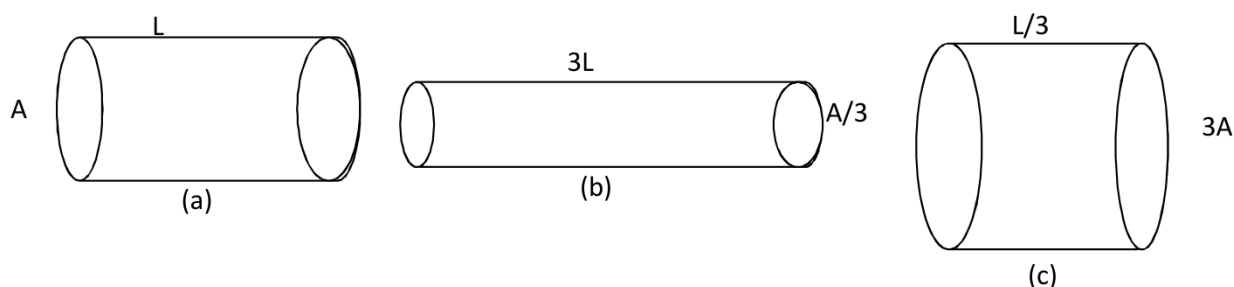
10. [3 marks] Source Q13 • 2018-19

13. The figure below shows three cylindrical copper conductors along with their face areas and lengths. Compare the resistance and the resistivity of the three conductors. Justify your answer.

3



3



Source/editorial note: Diagram labels verified visually: (a) L and A ; (b) $3L$ and $A/3$; (c) $L/3$ and $3A$. The full question crop preserves a stray printed "3" below the figures as present in the source.

11. [3 marks] Source Q7 • 2017-18

(a) Lights and fans: (b) Appliances of 2kW or more power:		
7	Draw a circuit diagram of an electric circuit containing a cell, a key, an ammeter, a resistor of $4\ \Omega$ in series with a combination of two resistors ($8\ \Omega$ each) in parallel and a voltmeter across parallel combination. Each of them dissipate maximum energy and can withstand a maximum power of 16W without melting. Find the maximum current that can flow through the three resistors.	3
8	In the electrolysis of water	2

Source/editorial note: Question asks the candidate to draw a circuit; no diagram is supplied in the SQP. Original wording "Each of them dissipate maximum energy" retained. In the connected circuit the $4\ \Omega$ resistor limits the total current to 2 A; each $8\ \Omega$ resistor then dissipates 8 W, not 16 W. The official answer follows this limiting-current interpretation. Final audit replaced clipped original crop; complete final "resistors." line visually confirmed.

12. [3 marks] Source Q37 VI • 2026-27

For visually impaired students

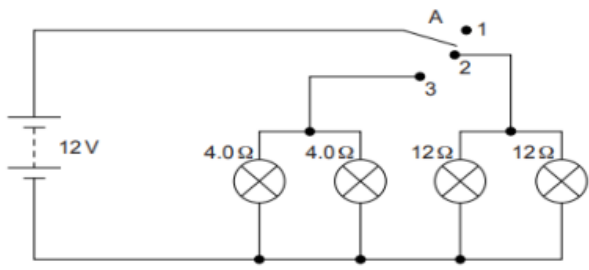
A 15 V battery is connected in series with three resistors of $2\ \Omega$, $3\ \Omega$, and $5\ \Omega$. Calculate:

- The total resistance in the circuit.
- The total current supplied by the battery.
- The voltage drop across the $3\ \Omega$ resistor.

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

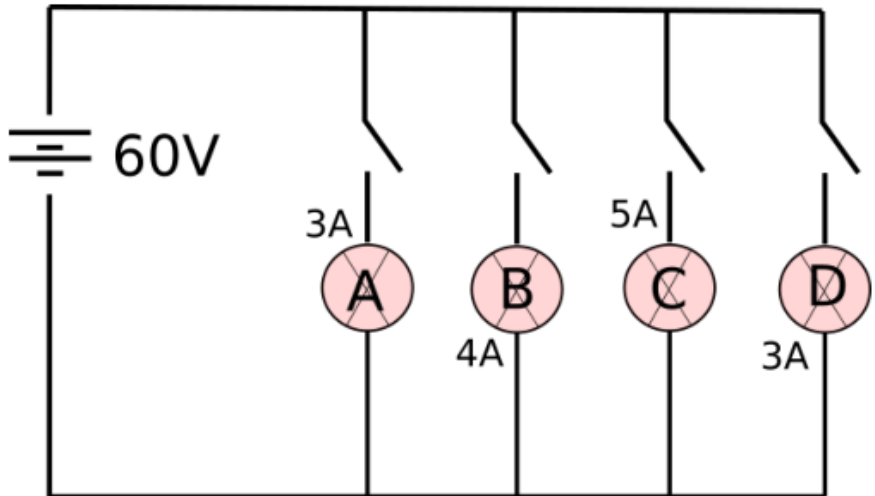
Section D — 5-mark / case-based questions

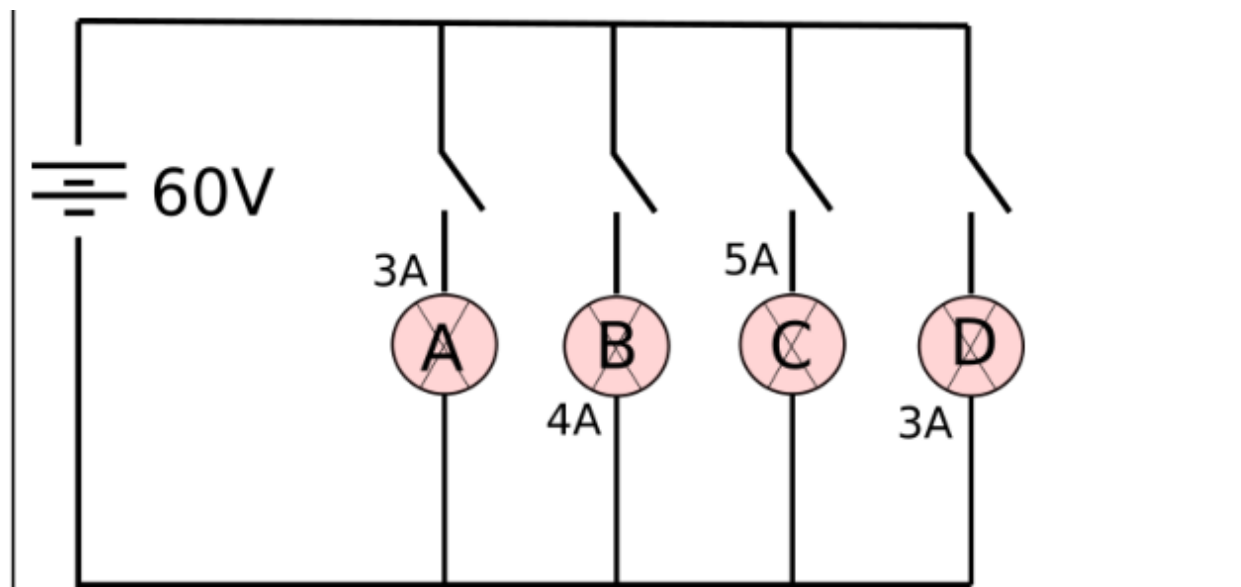
13. [4 marks] Source Q39 OR iv • 2023-24

39	 Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions. - State what happens when switch A is connected to - Position 2 - Position 3 - Find the potential difference across each lamp when lit. OR - Show, with calculations, which type of lamp, $4.0\ \Omega$ or $12\ \Omega$, has the higher power.	1+1+2
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Source/editorial note: Includes common (i), (ii) and OR (iv); 1+1+2 marks. Original scheme mislabels second line of (i) as (a). Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

14. [5 marks] Source Q36 • 2020-21

36	In the given circuit, A, B, C and D are four lamps connected with a battery of 60V.  Analyse the circuit to answer the following questions. (i) What kind of combination are the lamps arranged in (series or parallel)? (ii) Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps? (iii) Explain with proper calculations which lamp glows the brightest? (iv) Find out the total resistance of the circuit.	5
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Source/editorial note: All four subparts retained. The OR alternative on pp.14-15 concerns magnetic fields and the right-hand thumb rule and is excluded. Source MS p.8 prints " $= 60/15 \text{ A} = 4\text{A}$ "; the resistance unit is corrected explicitly to Ω , not silently attributed to the source.

Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — pyq-2026-a53ce36a0fd993043d965609

Official answer unavailable/withheld: No independently verified official answer supplied.

Original marking scheme (see limitation above): https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086_SCIENCE.zip

Question 2 — SQP_2026-27_q31

C. 4.0 A

31	C. 4.0 A	1
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Marking-scheme PDF page(s): [7]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

Question 3 — cbse_sqp_2019-20_electricity_q7

i) volt-ampere. Power = Voltage $\times$ Current.

7	i) volt-ampere Power = Voltage $\times$ Current.	1 mark
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

Question 4 — sqp_2022-23_q15

(b) diameter d of the wire

15.	(b) diameter d of the wire	1
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

Question 5 — pyq-2023-246eac3017f0793c27bba6a6

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 6 — pyq-2024-1e2d386b35acad1e9e07f369

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 7 — SQP_2024-25_q25_VI_B

B. $P = V \times I$ (0.5)

$I = P/V = 1000/220 = 4.54$ A (0.5)

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

OR

$$B. P = V \times I \quad (0.5)$$

$$I = \frac{P}{V} = \frac{1000}{220} = 4.54 \text{ A} \quad (0.5)$$

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

Marking-scheme PDF page(s): [3]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

Question 8 — SQP_2025-26_q34_A

A.

$$(i) R = R_1 R_2 / (R_1 + R_2) = (8 \times 8) / (8 + 8) = 4 \text{ ohms}$$

$$(ii) I = V/R = 8/(4+4) = 1 \text{ A}$$

34

Student to attempt either A or B.

2

A.

(i)

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{8 \times 8}{8 + 8} = 4 \text{ ohms}$$

(ii)

$$I = \frac{V}{R} = \frac{8}{(4 + 4)} = 1 \text{ A}$$

OR

Marking-scheme PDF page(s): [7]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

Question 9 — pyq-2025-74354640f102dee3b9237ff5

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 10 — cbse_sqp_2018-19_electricity_q13

$R_a = \rho L/A$; $R_b = \rho(3L)/(A/3) = 9\rho L/A = 9R_a$; $R_c = \rho(L/3)/(3A) = (1/9)\rho L/A = (1/9)R_a$. Hence $R_b > R_a > R_c$. $\rho_a = \rho_b = \rho_c$ because all three conductors are of the same material.

$$13. R_a = \rho \frac{L}{A}$$

3

$$R_b = \rho \left(\frac{3L}{A/3} \right) = 9 \frac{\rho L}{A} = 9 R_a$$

$$R_c = \rho \frac{L/3}{3A} = \frac{1}{9} \frac{\rho L}{A} = \frac{1}{9} R_a$$

Hence $R_b > R_a > R_c$

$\rho_a = \rho_b = \rho_c$ because all the three conductors are of same material.

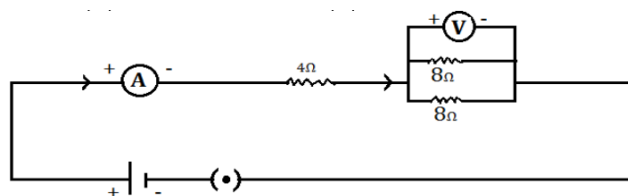
Marking-scheme PDF page(s): [4]

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

Question 11 — cbse_sqp_2017-18_electricity_q7

Circuit diagram as in MS page 1. Maximum current through 4 Ω resistor = $\sqrt{(P/R)} = \sqrt{(16/4)} = 2 \text{ A}$.
Maximum current through each 8 Ω resistor = $(1/2) \times 2 = 1 \text{ A}$.

7



1

Maximum current through 4 Ω resistor $= \sqrt{\frac{P}{R}}$

$$= \sqrt{\frac{16}{4}} = 2A$$

1

∴ Maximum current through each 8 Ω resistor $= \frac{1}{2} \times 2 = 1A$

1

Marking-scheme PDF page(s): [1, 2]

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

Question 12 — SQP_2026-27_q37_VI

A. Total Resistance:

$$R_{\text{total}} = 2 + 3 + 5 = 10 \Omega \text{ (1)}$$

B. Total current:

$$I = V/R_{\text{total}} = 15/10 = 1.5 \text{ A. (1)}$$

C. Voltage drop across the 3 Ω resistor:

$$V_{3\Omega} = I \times 3 = 1.5 \times 3 = 4.5 \text{ V (1)}$$

	<u>For visually impaired students</u>	
	A. Total Resistance: $R_{\text{total}} = 2 + 3 + 5 = 10 \Omega$ B. Total current: $I = \frac{V}{R_{\text{total}}} = \frac{15}{10} = 1.5 \text{ A.}$ C. Voltage drop across the 3 Ω resistor: $V_{3\Omega} = I \times 3 = 1.5 \times 3 = 4.5 \text{ V}$	(1) (1) (1)

Marking-scheme PDF page(s): [9, 10]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

Question 13 — SQP_2023-24_q39_OR_iv

(i) 12 Ω lamps (only) on.

(a) 4 Ω lamps (only) on [0.5+0.5]

(ii) 12 V for both sets of lamps and all of them are in parallel. [1]

OR

$$P = V^2/R$$

All lamps are in parallel and hence same V for all lamps.

$$\text{For } 4 \Omega \text{ lamps} \rightarrow P = (12 \times 12)/4 = 36 \text{ W}$$

$$\text{For } 12 \Omega \text{ lamps} \rightarrow P = (12 \times 12)/12 = 12 \text{ W}$$

Hence 4 Ω lamps will have higher power. [0.5 x 4]

39	(i) 12 Ω lamps (only) on. (a) 4 Ω lamps (only) on (ii) 12 V for both sets of lamps and all of them are in parallel.	[0.5+0.5] [1]	4
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OR

$$P = V^2 / R$$

All lamps are in parallel and hence same V for all lamps.

$$\text{For } 4 \Omega \text{ lamps} \rightarrow P = \frac{12 \times 12}{4} = 36 \text{ W}$$

$$\text{For } 12 \Omega \text{ lamps} \rightarrow P = \frac{12 \times 12}{12} = 12 \text{ W}$$

Hence 4 Ω lamps will have higher power.

[0.5 x 4]

Marking-scheme PDF page(s): [6]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

Question 14 — sqp_2020-21_q36_main

(i) The lamps are in parallel. (ii) A faulty lamp does not affect the working of the others; all lamps use the full potential of the battery. (iii) The lamp with the highest power glows brightest. All have the same voltage; C has the largest current. $PC = 5 \times 60 = 300 \text{ W}$, the maximum. (iv) Total current = $3 + 4 + 5 + 3 = 15 \text{ A}$; $R = V/I = 60/15 = 4 \Omega$. The MS prints the final unit incorrectly as A.

36	The lamps are in parallel. Advantages: If one lamp is faulty, it will not affect the working of the other lamps. They will also be using the full potential of the battery as they are connected in parallel.	1 1
	The lamp with the highest power will glow the brightest. $P=VI$ In this case, all the bulbs have the same voltage. But lamp C has the highest current. Hence, for Lamp C $P=5 \times 60 \text{ Watt} = 300 \text{ W}$. (the maximum).	1 1
	The total current in the circuit = $3+4+5+3 \text{ A} = 15\text{A}$ The Voltage = 60V $V=IR$ and hence $R = V/I$ $= 60/15 \text{ A} = 4\text{A}$	1

Marking-scheme PDF page(s): [8]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

Per-question source appendix

Original CBSE questions/images remain attributed to their source. Compilation selection and numbering are editorial. The full Science source PDFs may include unrelated chapters; only selected Electricity crops form this worksheet.

1. pyq-2026-a53ce36a0fd993043d965609

CBSE Class X Science 2026 First Board 31-3-2 | PYQ | Year/session: 2026 / 2026 First Board | Paper code: 31/3/2 | Source question: 31 | Original section: Section C (Physics) | Selected marks: 1 | PDF pages (one-based): 23

<https://www.cbse.gov.in/cbsenew/question-paper/2026/X/Science.zip>

Also used in: official-04, official-14

2. SQP_2026-27_q31

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 31 | Original section: Section – C | Selected marks: 1 | PDF pages (one-based): 11

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Also used in: official-01, official-05, official-11, official-12, official-15

3. cbse_sqp_2019-20_electricity_q7

Class-X Science-086 SAMPLE QUESTION PAPER 2019-20 | SQP | Year/session: 2019-20 / 2019-20 | Paper code: Not recorded | Source question: 7 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Also used in: official-02, official-04, official-12, official-14

4. sqp_2022-23_q15

CBSE Class X Science (086) Sample Question Paper 2022-23 | SQP | Year/session: 2022-23 / 2022-23 | Paper code: Not recorded | Source question: 15 | Original section: Section A | Selected marks: 1 | PDF pages (one-based): 5

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Also used in: official-02, official-04, official-10, official-12, official-14

5. pyq-2023-246eac3017f0793c27bba6a6

CBSE — Class X Science — 2023 — main — 31/2/1 | PYQ | Year/session: 2023 / main | Paper code: 31/2/1 | Source question: 25 | Original section: SECTION B | Selected marks: 2 | PDF pages (one-based): 15

<https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>

Also used in: official-04, official-07, official-11, official-14

6. pyq-2024-1e2d386b35acad1e9e07f369

CBSE — Class X Science — 2024 — main — 31/4/1 | PYQ | Year/session: 2024 / main | Paper code: 31/4/1 | Source question: 26 | Original section: SECTION B | Selected marks: 2 | PDF pages (one-based): 13

<https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>

Also used in: official-05, official-11, official-14

7. SQP_2024-25_q25_VI_B

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 25 VI B | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 7

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Also used in: official-03

8. SQP_2025-26_q34_A

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 34 A | Original section: Section - C | Selected marks: 2 | PDF pages (one-based): 11

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Also used in: official-05, official-12

9. pyq-2025-74354640f102dee3b9237ff5

CBSE — Class X Science — 2025 — main — 31/1/2 | PYQ | Year/session: 2025 / main | Paper code: 31/1/2 | Source question: 27 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 15

https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip

Also used in: official-05, official-15

10. cbse_sqp_2018-19_electricity_q13

Class-X Science-086 SAMPLE QUESTION PAPER-19 | SQP | Year/session: 2018-19 / 2018-19 | Paper code: Not recorded | Source question: 13 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 3

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Also used in: official-03, official-14

11. cbse_sqp_2017-18_electricity_q7

Sample Question Paper 2017-18 — Science — Class - X | SQP | Year/session: 2017-18 / 2017-18 | Paper code: Not recorded | Source question: 7 | Original section: SECTION - A | Selected marks: 3 | PDF pages (one-based): 1

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

Also used in: official-03

12. SQP_2026-27_q37_VI

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 37 VI | Original section: Section - C | Selected marks: 3 | PDF pages (one-based): 13

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

No cross-paper reuse.

13. SQP_2023-24_q39_OR_iv

CBSE Class X Science (086) Sample Question Paper 2023-24 (English) | SQP | Year/session: 2023-24 / 2023-24 | Paper code: Not recorded | Source question: 39 OR iv | Original section: Section-E | Selected marks: 4 | PDF pages (one-based): 8

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Also used in: official-01, official-11, official-12, official-15

14. sqp_2020-21_q36_main

CBSE Class X Science (086) Sample Question Paper 2020-21 | SQP | Year/session: 2020-21 / 2020-21 | Paper code: Not recorded | Source question: 36 | Original section: Section D | Selected marks: 5 | PDF pages (one-based): 14

https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Also used in: official-03, official-05, official-12